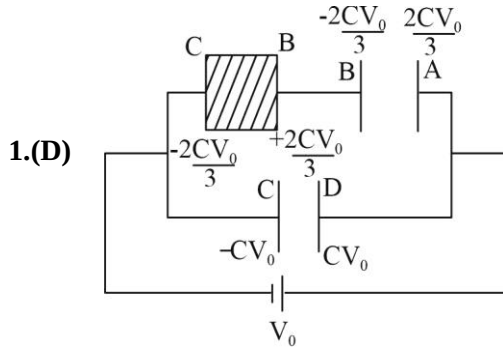


Solutions to JEE Advanced -5 | JEE 2022 | Paper-1

PHYSICS



$$V_{AB} = \frac{2V_0}{3}$$

$$V_{BC} = \frac{V_0}{3}$$

$$V_{DC} = V_0$$

2.(B) $i_{\max}$ (in steady state) $= \frac{M \frac{dI}{dt}}{R} = \frac{\mu_0 n \pi a^2 I_0}{R}$

Where $M = (\mu_0 n \pi a^2)$

3.(B) $R_{AB} = \alpha(1)^2 = 8 \Rightarrow \alpha = \frac{8\Omega}{m^2}$

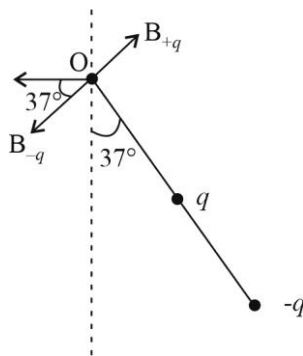
$$E_{eq} = \frac{2}{3}(25 - 10) = \left(\frac{50}{10}\right) \cdot 8x^2$$

$$\Rightarrow x^2 = \frac{1}{4} \text{ or } x = \frac{1}{2} = 0.5 \text{ m}$$

4.(A) $B_{+q} = \frac{\frac{\mu_0 q}{4\pi} \left(\frac{\ell}{2} \omega_0\right) (\sin 37^\circ)}{\left(\frac{\ell}{2}\right)^2}$

$$B_{-q} = \frac{\frac{\mu_0 q}{4\pi} (\ell \omega_0) (\sin 37^\circ)}{\ell^2}$$

$$B_{net} = \frac{\mu_0 q \omega_0 (\sin 37^\circ)}{4\pi \ell}$$



$$\begin{aligned}
 5.(B) \quad u &= \frac{3KQ_0^2}{5a} + \frac{3K(2Q_0)^2}{\frac{5a}{2}} + \left(-\frac{KQ_0(2Q_0)}{2a} \right) \\
 &= \frac{KQ_0^2}{a} \left[\frac{3}{5} + \frac{24}{5} - 1 \right] = \frac{KQ_0^2}{a} \left[\frac{22}{5} \right] = \frac{11Q_0^2}{10\pi\epsilon_0 a}
 \end{aligned}$$

$$6.(D) \quad \vec{dA} = dA \frac{x\hat{i} + y\hat{j}}{\sqrt{x^2 + y^2}}$$

$$d\phi = 0 \text{ as } \vec{E} \perp \vec{dA}$$

So net flux is also zero

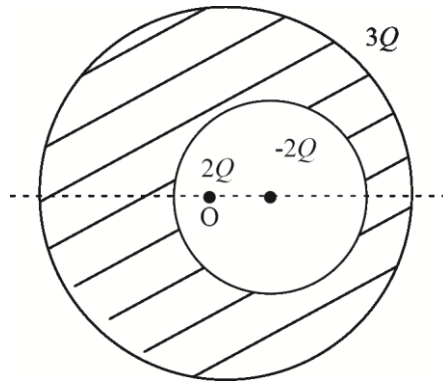
$$7.(AC) \quad F_{elec} = qE = \frac{e(nlA)enA}{2\pi\epsilon_0 r} = \frac{e^2 n^2 A^2 l}{2\pi\epsilon_0 r}$$

$$F_{mag} = \frac{\mu_0 i^2 l}{2\pi r} = \frac{\mu_0 e^2 n^2 v^2 A^2 l}{2\pi r}$$

$$\frac{F_{mag}}{F_{elec}} = v^2 \mu_0 \epsilon_0 = \frac{v^2}{c^2}$$

Beams will diverge $F_{elec} \gg F_{mag}$.

8.(BC)



$$E_{(3R)} = \frac{K(3Q)}{9R^2}$$

$$V_0' = \frac{K(2Q)}{R/4} - \frac{K(2Q)}{R/2} + \frac{K(3Q)}{R} = \frac{KQ}{R}(8 - 4 + 3) = \frac{7KQ}{R} = \frac{7Q}{4\pi\epsilon_0 R}$$

$$\text{At P } \vec{E}_{2Q} + \vec{E}_{induced} = 0$$

$$\vec{E}_{induced} = \frac{K(2Q)}{(3R)^2} = \frac{2KQ}{9R^2}$$

If a charge is brought from ∞ and placed outside, field outside the sphere will change so potential of point O' will also change.

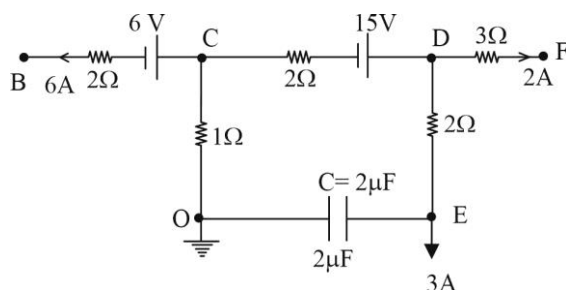
9.(ABC) $V_C = 0 - 11 = -11V$

$V_B = -11 - 6 - 12 = -29V$

$V_D = -11 - 10 + 15 = -6V$

$V_E = -6 - 6 = -12V$

$V_{\text{capacitor}} = \frac{1}{2} \times 2 \times 10^{-6} \times 12^2 = 144 \mu J$



10.(BD) $\epsilon_{\text{ind}} = Bv_0 d$ and electric field between plates $E = Bv_0$

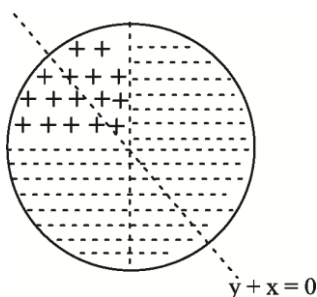
$\frac{\sigma}{\epsilon_0} = Bv_0$ or $\sigma = Bv_0 \epsilon_0$

- Current $i = \frac{Bv_0 d}{R + \frac{\rho d}{A}}$

- P_{max} when $R = \frac{\rho d}{A}$

- $P_{\text{max}} = \frac{(Bv_0 d)^2}{4 \frac{\rho d}{A}} = \frac{B^2 v_0^2 d A}{4 \rho}$

11.(CD)



Electric field at a point on y-axis will make an acute angle with -ve y axis.

Electric field at any point on line $y = -x$ will be along the line.

At any point on z-axis. (0, 0, R)

$E_z = \frac{-\sigma}{4\epsilon_0} \left[1 - \frac{R}{\sqrt{2}R} \right] = \frac{-\sigma}{4\epsilon_0} \left[1 - \frac{1}{\sqrt{2}} \right]$

$W = q \int E_z dz = -q \frac{\sigma}{4\epsilon_0} \int_R^{2R} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right) dz = -q \frac{\sigma}{4\epsilon_0} \left[R + (\sqrt{2} - \sqrt{5})R \right] = \frac{Rq\sigma}{4\epsilon_0} [\sqrt{5} - \sqrt{2} - 1]$

12.(ABC) $\int_{\rho_0}^{\rho} \frac{d\rho}{\rho} = \int_{T_0}^T \alpha dT \Rightarrow \rho = \rho_0 e^{\alpha(T-T_0)} = \rho_0 (1 + \alpha(T-T_0))$ if $\alpha(T-T_0) \ll 1$

$R = \rho \frac{l}{A}$ so with increase in temperature.

$R = \frac{R_0 [1 + \alpha(T-T_0)] [1 + \alpha'(T-T_0)]}{1 + 2\alpha'(T-T_0)}$

With increase in temperature number density of conduction electrons increases.

1.(30) 2.(10)

$$q = 4 + 3\sin(2\pi t) \Rightarrow i = 6\pi \cos(2\pi t)$$

$$P_{av} = \frac{\int_{0.5}^{0.75} (4 + 3\sin(2\pi t)) 6\pi \cos(2\pi t) dt}{1/4}$$

$$= 4 \left[\int_{0.5}^{0.75} 24\pi \cos(2\pi t) dt + 9\pi \int_{0.5}^{0.75} \sin(4\pi t) dt \right] = 4 \left[\frac{24\pi(-1)}{2\pi} + \frac{9\pi}{4\pi} (+2) \right]$$

$$= \frac{4}{4\pi} [-48 + 18] \pi = -30 \text{ watts}$$

$$i = \frac{4}{R} + \frac{3}{\sqrt{R^2 + X_C^2}} \sin(\omega t + \phi) = \frac{4}{R} + \frac{6}{4R} \sin(\omega t + \phi)$$

3.(2) 4.(5)

$$a = \frac{q\sqrt{(vB)^2 + E^2}}{m} = \frac{v^2}{R}$$

$$R = \frac{mv^2}{q\sqrt{(vB)^2 + E^2}} = \frac{9}{\sqrt{\frac{65}{4} + 4}} = 2m$$

$$W_E = q \left[\int_0^2 2x dx + \int_0^2 2dx \right] = \frac{1}{2} m (v^2 - 3^2)$$

$$4 + 4 = \frac{v^2 - 9}{2} \Rightarrow v^2 = 25 \Rightarrow v = 5m/s$$

5.(10) 6.(2)

i is maximum.

$$V_L = 0 \Rightarrow V_{1\mu F} + V_{2\mu F} = 0$$

$$V_{common} = \frac{20 \times 2 - 10 \times 1}{3} = 10 \text{ volt}$$

$$\frac{1}{2}(c)v^2 + \frac{1}{2}(2c)(2v)^2 = \frac{1}{2}(c + 2c)v_{comon}^2 + \frac{1}{2}Li^2$$

$$i = v\sqrt{\frac{6C}{L}} = 10\sqrt{\frac{6 \times 10^{-6}}{0.15 \times 10^{-3}}} = 10\sqrt{4 \times 10^{-2}} = 2 \text{ Amp}$$

CHEMISTRY

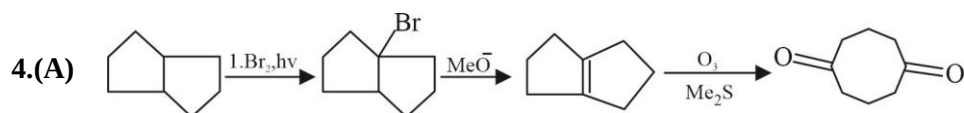
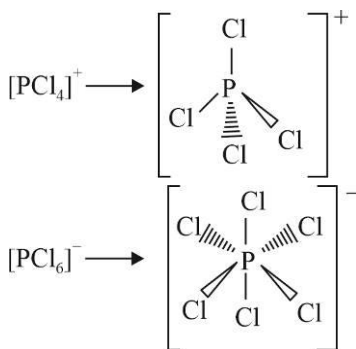
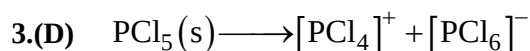
$$1.(C) \quad \frac{8}{3} = \frac{4 \times \text{Molar mass}}{(5 \times 10^{-8})^3 \times (6 \times 10^{23})}$$

Molar mass = 50 g per mol

At 0°C, the substance will exist as gas and its density

$$= \frac{50}{22.4} = 2.23 \text{ g / litre}$$

2.(B) In a minimum boiling azeotrope, the new interactions A – B are weaker than the existing A – A or B – B interactions.



5.(C) Sucrose is a non-reducing sugar and comprises $\text{C}_1 - \text{C}_2$ glycosidic linkage

Maltose is a reducing sugar and comprises $\text{C}_1 - \text{C}_4$ glycosidic linkage

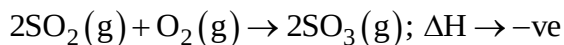
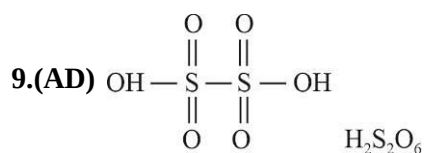
Lactose is a reducing sugar and upon hydrolysis gives β -D galactose and β -D glucose

Fructose gives positive tollen's test

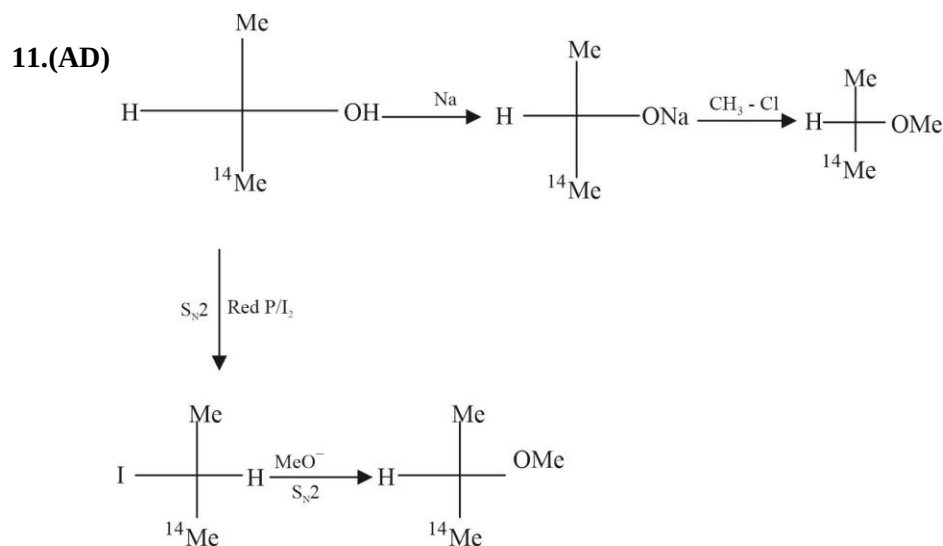
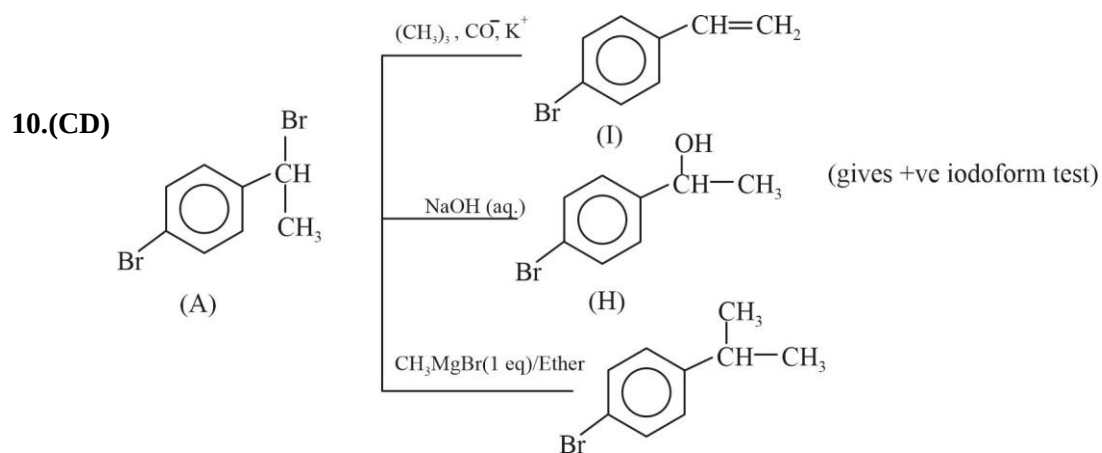
6.(C) In α -helix structure of protein, a polypeptide chain is stabilized by the formation of intramolecular H-bonds between $-\text{NH}-$ group of amino acids in one turn with the $\text{C}=\text{O}$ groups of amino acids belonging to adjacent turn.

7.(AB) Van't Hoff's factor 'i' for X will be more than one while it will be less than one for Y. Hence B.P. ; osmotic pressure (π) and lowering in V.P. will be more in X in comparison to Y.

8.(ACD) When KCl is heated in vapour of K, some of the Cl^- leave their lattice site and create anion vacancies. This chloride ion combines with K vapour to form potassium chloride. In doing so K atom loses electrons to form K^+ ions. This released electron diffuses into the crystal to get entrapped in the anion vacancy called F-centre.



Preparation of NH_3 by Haber's process is exothermic in nature.



Conversion of 'a' to I, and 'b' to II obey $\text{S}_{\text{N}}2$ mechanism.

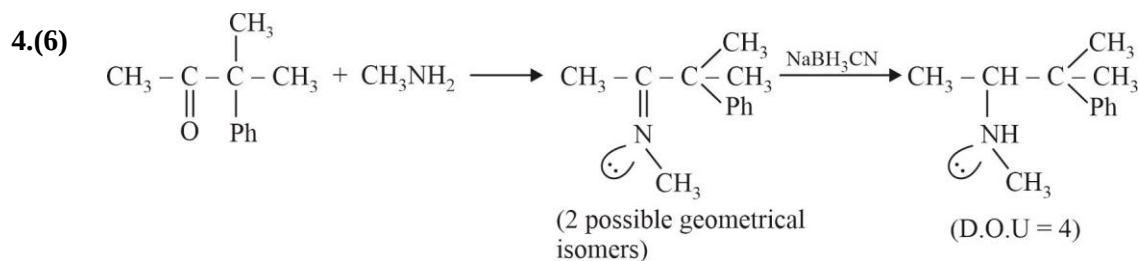
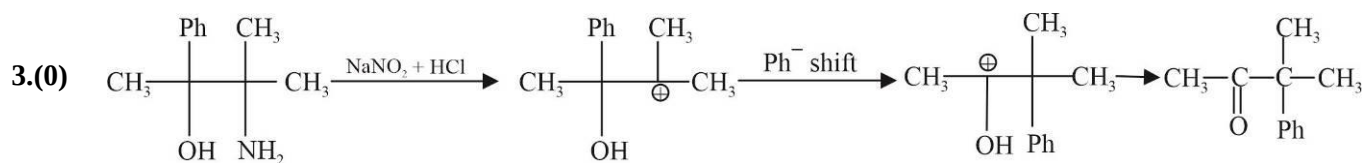
12.(ABC) The distance between two A^+ ions is $= \sqrt{\left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2}$
 $= \frac{a}{\sqrt{2}} = \frac{2(r_{\text{A}} + r_{\text{B}})}{\sqrt{2}} = \sqrt{2}(r_{\text{A}} + r_{\text{B}})$

1.(4) This packing refers to ABCABC type. Which is observed in FCC system.

Effective number of atoms in FCC unit cell = 4

2.(4) Distance between the centres of the balls A and D is equal to the distance of body diagonal in FCC packing. It is equal to $\sqrt{3}a = \sqrt{3} \times 2\sqrt{2}r = \sqrt{3} \times 2\sqrt{2} \times 2\text{cm} = 4\sqrt{6}\text{cm}$

$$\frac{\ell}{\sqrt{6}} = 4$$



$$x + y = 6$$

5.(6) 6.(6)

From lowering of vapour pressure information :

$$\frac{0.3}{50.3} = \chi_2 = \frac{n_2}{n_1 + n_2}$$

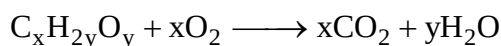
$$\Rightarrow \frac{n_1}{n_2} + 1 = \frac{503}{3}$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{500}{3}$$

$$\Rightarrow \frac{1000}{18} \times \frac{M}{50} = \frac{500}{3}$$

$$\Rightarrow M = 150 \text{ g/mol}$$

Also, the combustion reaction is :



$\therefore$ 18y g of H_2O is produced from 1.0 mole of compound

$$\therefore 0.9 \text{ g of } \text{H}_2\text{O} \text{ will be produced from } \frac{0.9}{18y} = \frac{1}{20y} \text{ mol}$$

$$\Rightarrow \text{At the end, moles of } \text{O}_2 \text{ left} = \frac{x}{20y}$$

$$\text{Moles of } \text{CO}_2 \text{ formed} = \frac{x}{20y}$$

$$\Rightarrow \text{Total moles of gases at STP} = \frac{2x}{20y} = \frac{2.24}{22.4}$$

$$\Rightarrow x = y$$

$$\Rightarrow \text{Molar mass ; } 150 = 12x + 2x + 16x = 30x$$

$$\Rightarrow x = \frac{150}{30} = 5$$

$$\Rightarrow \text{Formula} = \text{C}_5\text{H}_{10}\text{O}_5$$

MATHEMATICS

1.(D) Given $f(3x) = f(x)$

$$\therefore f(3x) = f(x) = f\left(\frac{x}{3}\right) = \dots = f\left(\frac{x}{3^n}\right)$$

As $n \rightarrow \infty; 3^n \rightarrow \infty$

$$\therefore \frac{x}{3^n} \rightarrow 0$$

$$\therefore f(3x) = f(0) \Rightarrow f(x) \text{ is a constant function}$$

$$\therefore f(x) = 5 \quad \forall x \in R$$

$$\operatorname{sgn}(f(x)) = 1$$

$$\int_2^5 \operatorname{sgn}(f(x)) dx = \int_2^5 dx = 3$$

2.(C) $f^4(x) = \int_0^x t^2 f^3(t) dt$ differentiating both sides w.r.t. x ,

$$4f^3(x) f'(x) = x^2 f^3(x) \Rightarrow f'(x) = \frac{x^2}{4}$$

Integrate both sides :

$$f(x) = \frac{x^3}{12} + C, \text{ since } f(0) = 0$$

$$\therefore C = 0$$

$$\therefore f(x) = \frac{x^3}{12}$$

$$A = \left| \int_{-1}^0 \frac{x^3}{12} dx \right| + \left| \int_0^4 \frac{x^3}{12} dx \right| \Rightarrow A = \left| \frac{x^4}{48} \right|_{-1}^0 + \left| \frac{x^4}{48} \right|_0^4 \Rightarrow A = \frac{1}{48} + \frac{4^4}{48} = \frac{257}{48}$$

$$3.(B) \text{ Sum} = \sum_{n=1}^{\infty} \tan^{-1} \left(\frac{2n+1}{n^4 + 2n^3 + n^2 + 1} \right) = \sum_{n=1}^{\infty} \tan^{-1} \left\{ \frac{(n+1)^2 - n^2}{1 + (n+1)^2 n^2} \right\}$$

$$= \sum_{n=1}^{\infty} \left(\tan^{-1}(n+1)^2 - \tan^{-1} n^2 \right)$$

$$= \lim_{n \rightarrow \infty} \left\{ \left(\tan^{-1} 2^2 - \tan^{-1} 1 \right) + \left(\tan^{-1} 3^2 - \tan^{-1} 2^2 \right) \right.$$

$$\left. + \left(\tan^{-1} 4^2 - \tan^{-1} 3^2 \right) + \dots + \left(\tan^{-1}(n+1)^2 - \tan^{-1} n^2 \right) \right\}$$

$$= \lim_{n \rightarrow \infty} \left(\tan^{-1}(n+1)^2 - \tan^{-1} 1 \right) = \frac{\pi}{2} - \tan^{-1} 1 = \frac{\pi}{4}$$

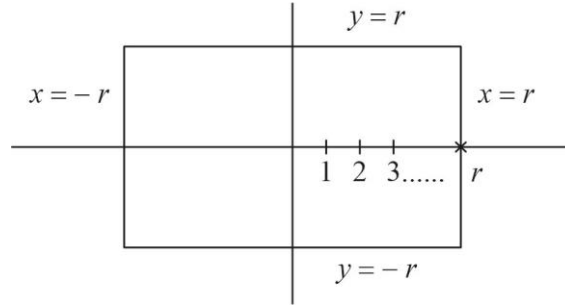
4.(A) $P(r) = (r+1)^2 + 2r(r+1) + r^2$

$$\sum_{r=1}^{n-1} \sqrt{P(r)} = \sum_{r=1}^{n-1} (2r+1) = n^2 - 1$$

$$\therefore \frac{1}{2} = \lim_{n \rightarrow \infty} \frac{n^2 - 1}{an^3 + bn^2 + c}$$

$$\Rightarrow a = 0 \quad \frac{1}{b} = \frac{1}{2} \quad \Rightarrow \quad b = 2$$

$$\therefore a + b = 2$$



5.(A) $f(0^+) = \lim_{x \rightarrow 0^+} |x|^{\sin x} = e^{\lim_{x \rightarrow 0} \sin x \ln x} = e^{\lim_{x \rightarrow 0} \frac{\ln x}{\operatorname{cosec} x}} = e^0 = 1$

$$f(0^-) = g(0) = 1$$

Here $g(x) = ax + b$

So $g(0) = b = 1$ thus $g(x) = ax + 1$

For $x > 0$, $f'(x) = e^{\sin x \ln x} \left[\cos x \ln x + \frac{\sin x}{x} \right]$

$$f'(1) = 1(0 + \sin 1) = \sin 1$$

$$f(1) = g(-1) = -a + 1$$

So $-a + 1 = \sin 1$ thus $a = 1 - \sin 1$

$$g(x) = (1 - \sin 1)x + 1$$

$$g(-101) = -101 + 101\sin 1 + 1 = 101\sin 1 - 100$$

6.(D) Domain $1 + \frac{1}{x} > 0 \Rightarrow \frac{(x+1)}{x} > 0$

So domain $(-\infty, -1) \cup (0, \infty)$

$$f(x) = e^{x \ln \left(\frac{x+1}{x} \right)}$$

$$f'(x) = e^{x \ln \left(1 + \frac{1}{x} \right)} \left[1 \ln \left(1 + \frac{1}{x} \right) + \frac{x}{1} \left(\frac{1}{x+1} - \frac{1}{x} \right) \right]$$

Let $g(x) = \ln \left(1 + \frac{1}{x} \right) - \frac{1}{(x+1)}$

$$g'(x) = \frac{1}{1+x} - \frac{1}{x} + \frac{1}{(x+1)^2} = -\frac{1}{x(x+1)} + \frac{1}{(x+1)^2} = \frac{-(x+1) + x}{x(x+1)^2} = -\frac{1}{x(x+1)^2}$$

(i) for $x \in (0, \infty)$, $g'(x) < 0$

$$g(x) > \lim_{x \rightarrow \infty} g(x) \quad \text{so} \quad g(x) > 0 \quad \text{so} \quad f'(x) > 0$$

(ii) for $x \in (-\infty, -1)$, $g'(x) > 0$

$$g(x) > \lim_{x \rightarrow -\infty} g(x) \quad \text{so} \quad g(x) > 0 \quad \text{so} \quad f'(x) > 0$$

So $f(x)$ is monotonically increasing in its domain.

7.(ACD)

$$\lim_{x \rightarrow 0} f(x) \text{ DNE}$$

LHL at $x = \pi$

$$\lim_{x \rightarrow \pi^-} \frac{\cos^{-1}(\cos(\sin x)) - |x - \pi|}{\sin^3 x} = \lim_{x \rightarrow \pi^-} \frac{\sin x + (x - \pi)}{\sin^3 x} \quad (\because \sin x > 0 \text{ as } x \rightarrow \pi^-)$$

Let $x - \pi = t$

$$\lim_{t \rightarrow 0^-} \frac{-\sin t + t}{-\sin^3 t} = \frac{-1}{6}$$

RHL at $x = \pi$

$$\lim_{x \rightarrow \pi^+} \frac{-\sin x - (x - \pi)}{\sin^3 x} \quad (\because \sin x < 0 \text{ as } x \rightarrow \pi^+)$$

$$\text{Put } x - \pi = t = -\lim_{t \rightarrow 0} \frac{\sin t + t}{\sin^3 t} = \frac{1}{6} \quad \therefore \quad \text{Jump of discontinuity} = \frac{1}{6} - \left(\frac{-1}{6}\right) = \frac{1}{3}$$

8.(ABCD)

Putting $y = x$ in given functional equation

$$f(x) + f(x) = \frac{1}{x} + \frac{1}{x} \Rightarrow f(x) = \frac{1}{x}$$

$$\therefore \int_2^3 \frac{3(f(x))^5 - f(x)}{1 - (f(x))^4} dx = \int_2^3 \frac{\left(\frac{3}{x^5} - \frac{1}{x}\right)}{\left(1 - \frac{1}{x^4}\right)} dx = \int_2^3 \frac{\left(\frac{3}{x^7} - \frac{1}{x^3}\right)}{\left(\frac{1}{x^2} - \frac{1}{x^6}\right)} dx$$

$$\text{Let } \frac{1}{x^2} - \frac{1}{x^6} = t \Rightarrow \left(\frac{6}{x^7} - \frac{2}{x^3}\right) dx = dt$$

$$\text{When } x = 2, t = \frac{1}{4} - \frac{1}{64} = \frac{15}{64} \text{ and } x = 3, t = \frac{1}{9} - \frac{1}{729} = \frac{80}{729}$$

$$\therefore \text{Integral} = \frac{1}{2} \int_{15/64}^{80/729} \frac{dt}{t} = \frac{1}{2} (\ln t)_{15/64}^{80/729} = \frac{1}{2} \ln \left(\frac{80}{729} \cdot \frac{64}{15} \right) = \frac{1}{2} \ln \left(\frac{2^{10}}{3^7} \right) \equiv \frac{1}{2} \log \frac{2^\alpha}{3^\beta}$$

$$\therefore \alpha = 10, \beta = 7$$

9.(BCD)

$$(A) \quad \frac{1}{\sqrt{1-0}} < \frac{1}{\sqrt{1-x^{2n}}} < \frac{1}{\sqrt{1-x^2}} \quad \forall x \in (0, 1)$$

$$\Rightarrow \int_0^{0.5} 1 dx < \int_0^{0.5} \frac{dx}{\sqrt{1-x^{2n}}} < \int_0^{0.5} \frac{1}{\sqrt{1-x^2}} dx$$

$$\frac{1}{2} < \int_0^{0.5} \frac{dx}{\sqrt{1-x^{2n}}} < \sin^{-1} \frac{1}{2} = \frac{1}{2} < \int_0^{1/2} \frac{1}{\sqrt{1-x^{2n}}} dx < \frac{\pi}{6}$$

$$(B) \quad \frac{dy}{dx} = \frac{a}{x} + 2bx + 1$$

$$x_1 = 1 \text{ is extremum point} \Rightarrow a + 2b + 1 = 0 \quad \dots\dots (i)$$

$$x_2 = 2 \text{ is also extremum point} \Rightarrow \frac{a}{2} + 4b + 1 = 0 \quad \dots\dots (ii)$$

From (i) & (ii)

$$a = -\frac{2}{3} \quad b = -\frac{1}{6}$$

$$(C) \quad y = \cos^2\left(x + \frac{\pi}{3}\right) - \sin^2 x + 1 - \cos x \cos\left(\frac{\pi}{3} + x\right)$$

$$= \cos\left(2x + \frac{\pi}{3}\right) \cos \frac{\pi}{3} + 1 - \frac{1}{2} \left(\cos\left(\left(2x + \frac{\pi}{3}\right) + \cos \frac{\pi}{3}\right) \right) = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\therefore \frac{dy}{dx} = 0$$

$$(D) \quad f(-x) = \int_0^{-x} \sqrt{1-t^4} dt$$

$$t = -v = \int_0^x -\sqrt{1-v^2} dv = \int_0^x -\sqrt{1-t^4} dt = -f(x)$$

$$\Rightarrow f(x) \text{ is odd function}$$

10.(ABCD)

$$f(x) = x^{1/3}(x-1)$$

$$f'(x) = \frac{1}{3x^{2/3}}(4x-1)$$

$$\therefore f(x) \text{ is increasing for } x > \frac{1}{4} \text{ and decreasing for } x < \frac{1}{4}$$

$$\text{Local minima at } x = \frac{1}{4}$$

$$f'(x) = \frac{4}{3}x^{1/3} - \frac{1}{3}x^{-2/3}$$

$$f'(x) \text{ does not exist at } x = 0$$

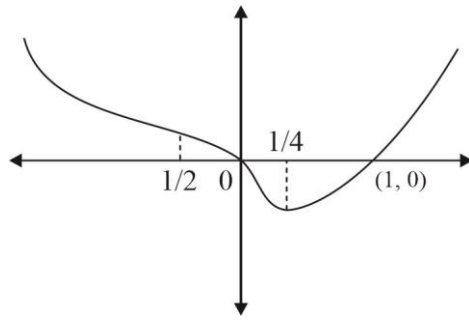
$$f'(0^-) \text{ and } f'(0^+) \text{ both tends to } \infty$$

$$\therefore \text{vertical tangent at } x = 0$$

$$f''(x) = \frac{2}{9x^{5/3}} \left(\frac{2x+1}{x} \right)$$

$$f''(x) = 0 \text{ at } x = \frac{-1}{2} \text{ (inflection point)}$$

Graph of $f(x)$ is



$$A = \int_{-1}^0 (x^{4/3} - x^{1/3}) dx + \left| \int_0^1 (x^{4/3} - x^{1/3}) dx \right|$$

$$A = \left(\frac{3}{7} x^{7/3} - \frac{3}{4} x^{4/3} \right) \Big|_{-1}^0 + \left| \left(\frac{3}{7} x^{7/3} - \frac{3}{4} x^{4/3} \right) \Big|_0^1 \right|$$

$$A = 0 - \left(-\frac{3}{7} - \frac{3}{4} \right) + \left| \frac{3}{7} - \frac{3}{4} \right| \Rightarrow A = \frac{12+21}{28} + \frac{9}{28} \Rightarrow A = \frac{42}{28} = \frac{3}{2}$$

11.(ABD)

$$f(x) = \int_0^x |t-1| dt,$$

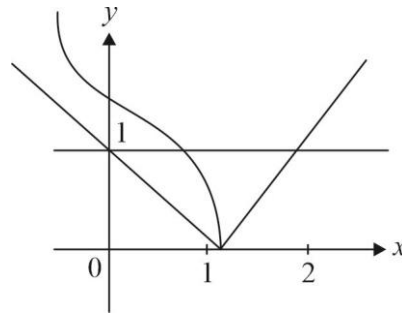
$$\Rightarrow f(x) = \int_0^x (1-t) dt, \quad 0 \leq x \leq 1 \quad = x - \frac{x^2}{2}$$

$$\text{Also, } f(x) = \int_0^1 (1-t) dt + \int_1^x (t-1) dt, \quad \text{Where } 1 \leq x \leq 2$$

$$= \frac{1}{2} + \frac{x^2}{2} - x + \frac{1}{2} = \frac{x^2}{2} - x + 1$$

$$f(x) = \begin{cases} x - \frac{x^2}{2} & ; 0 \leq x \leq 1 \\ \frac{x^2}{2} - x + 1 & ; 1 < x \leq 2 \end{cases}$$

$$f'(x) = \begin{cases} 1-x & 0 \leq x \leq 1 \\ x-1 & 1 < x \leq 2 \end{cases}$$

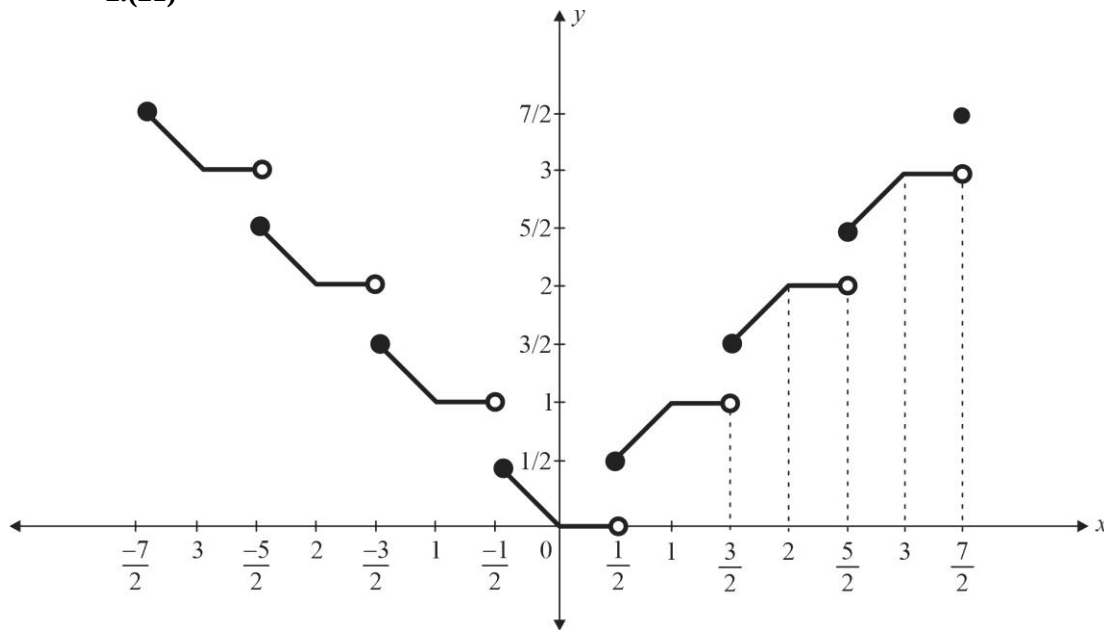


$$12.(CD) \tan^{-1} \left(\frac{2x}{1-x^2} \right) = \begin{cases} \pi + 2 \tan^{-1} x & , x < -1 \\ 2 \tan^{-1} x & , -1 < x < 1 \\ -\pi + 2 \tan^{-1} x & , x > 1 \end{cases}$$

$$2 \tan^{-1} x - \tan^{-1} \left(\frac{2x}{1-x^2} \right) = \begin{cases} -\pi & , x < -1 \\ 0 & , -1 < x < 1 \\ \pi & , x > 1 \end{cases}$$

1.(2)

2.(21)



$$L = 7, M = 14 \quad \therefore (L + M) = 21$$

3.(3.50)

4.(50)

$$f'(x) = 3a^2x^2 + 2ax - 1 = 0 \text{ at } x = -1$$

$$3a^2 - 2a - 1 = 0 \Rightarrow a = -\frac{1}{3}, 1$$

Since extreme values are positive

$$\therefore a = 1 \Rightarrow f(x) = x^3 + x^2 - x + b \quad f(-1) > 0$$

$$-1 + 1 + 1 + b > 0 \Rightarrow b > -1 \quad \therefore S_1 = \{0, 1, 2, 3, \dots\}$$

$$a = -\frac{1}{3} \quad f(x) = \frac{x^3}{9} - \frac{x^2}{3} - x + b \quad f(-1) > 0$$

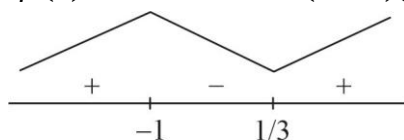
$$\Rightarrow \frac{-1}{9} - \frac{1}{3} + 1 + b > 0 \Rightarrow b > -\frac{5}{9} \quad S_2 = \{0, 1, 2, 3, \dots\}$$

$$S = \sum_{n \in S_1} \left(\frac{\cos \theta}{2} \right)^n + \sum_{p \in S_2} \left(\frac{\cos(\theta/2)}{3} \right)^p$$

$$= \frac{2}{2 - \cos \theta} + \frac{3}{3 - \cos(\theta/2)} \quad S_{\max} = 2 + \frac{3}{2} = \frac{7}{2} \text{ at } \theta = 4n\pi$$

$$\text{If } a = 1, b = 2 \quad f(x) = x^3 + x^2 - x + 2$$

$$f'(x) = 3x^2 + 2x - 1 = (3x - 1)(x + 1)$$



$$\therefore g(0) = f(-1) = 3$$

$$g(5) = f\left(\frac{1}{3}\right) = \frac{1 + 3 - 9 + 54}{27} = \frac{49}{27} \quad \therefore \frac{1}{3}g(0) + 27g(5) = 50$$

5.(1)

6.(5)

$$\begin{bmatrix} a^2 f(-1) + 4a f(1) + f(2) & 4a f(0) + 1 \\ b^2 f(-1) + 4b f(1) + f(2) & 4b f(0) + 1 \\ c^2 f(-1) + 4c f(1) + f(2) & 4c f(0) + 1 \end{bmatrix} = \begin{bmatrix} 2a^2 + 16a + 17 & 4a + 1 \\ 2b^2 + 16b + 17 & 4b + 1 \\ 2c^2 + 16c + 17 & 4c + 1 \end{bmatrix}$$

$x^2 (f(-1) - 2) + (f(1) - 4)x + f(2) - 17 = 0$ has 3 real and distinct roots a, b, c

$$\Rightarrow f(-1) = 2, f(1) = 4, f(2) = 17$$

Also $4x f(0) + 1 = 4x + 1$ has 3 real and distinct roots $\Rightarrow f(0) = 1$

Let $f(x) = ax^3 + bx^2 + cx + d$

$$f(0) = d = 1$$

$$\begin{aligned} -a + b - c + 1 &= 2 \\ a + b + c + 1 &= 4 \end{aligned} \Rightarrow b = 2$$

$$8a + 4b + 2c = 16$$

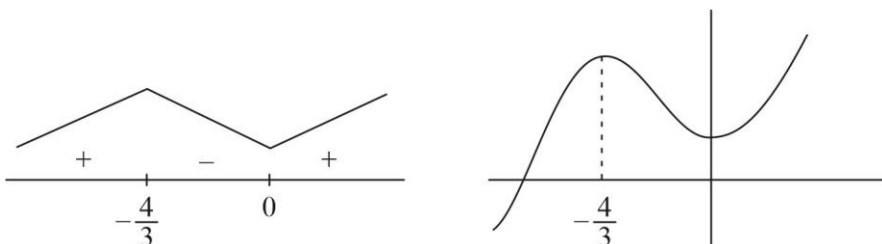
$$a + c = 1$$

$$4a + c = 4$$

$$3a = 3 \Rightarrow a = 1, c = 0$$

$$f(x) = x^3 + 2x^2 + 1$$

$$f'(x) = x(3x + 4)$$



Clearly least value in $[-1, 1]$ occurs at $x = 0$

$$m = f(0) = 1$$

$$f(1) = 4$$

$$f(-1) = 2$$

$$\therefore M = 4 \Rightarrow \text{Sum} = 5$$